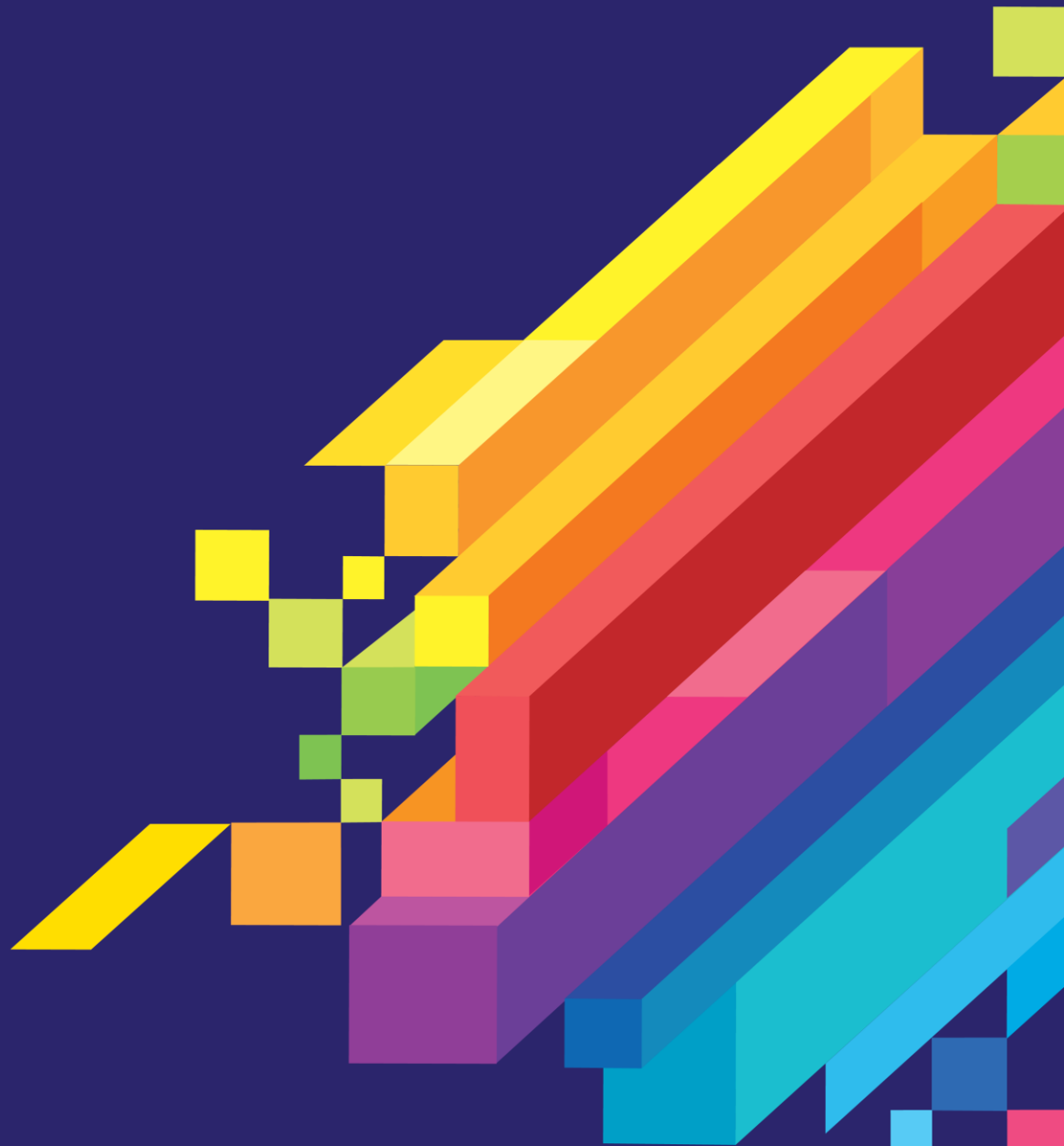
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Physical Layer

Topic 05 – *DLL and Physical Layer II*



Our vision is to be Singapore's private education institution of choice, and we are committed to helping our students achieve their career and educational goals through lifelong learning opportunities.

We build futures, one success story at a time.

Integrity . Knowledge . Support . Opportunity . Results

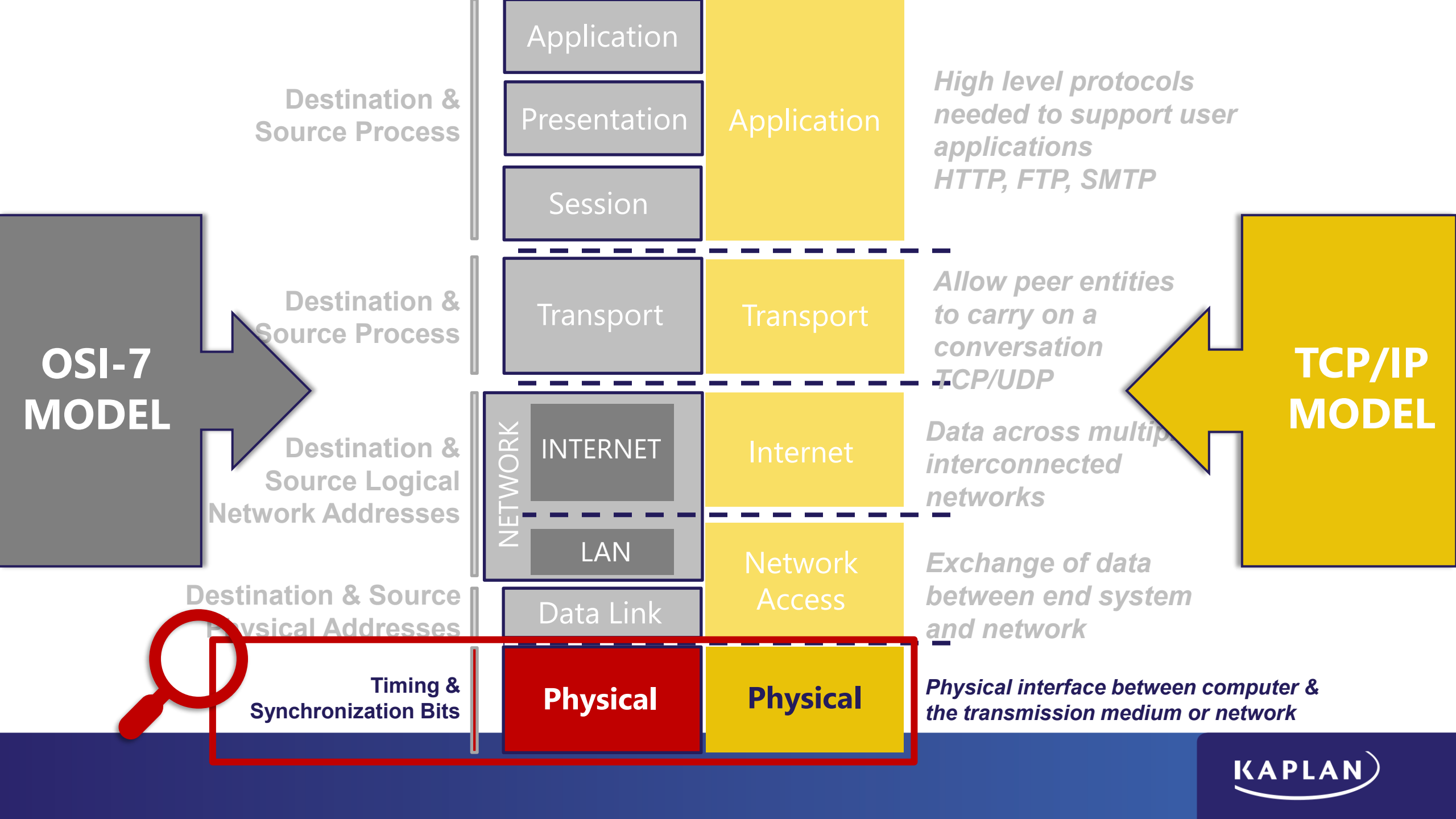
Topic 5 – Learning Outcomes

Network Topologies

-Physical Topology

-Logical Topology

-Hybrid Topology


OSI-7 MODEL
TCP/IP MODEL
Application
Presentation
Session
Application
Destination & Source Process
High level protocols needed to support user applications HTTP, FTP, SMTP
Transport
Transport
Destination & Source Process
Allow peer entities to carry on a conversation TCP/UDP
NETWORK
INTERNET
Internet
Destination & Source Logical Network Addresses
Data across multip interconnected networks
LAN
Network Access
Exchange of data between end system and network
Data Link
Destination & Source Physical Addresses
Timing & Synchronization Bits
Physical
Physical
Physical interface between computer & the transmission medium or network

Network Topology

- Refers to **how network devices are connected**
- 3 Aspects

- Physical Topology

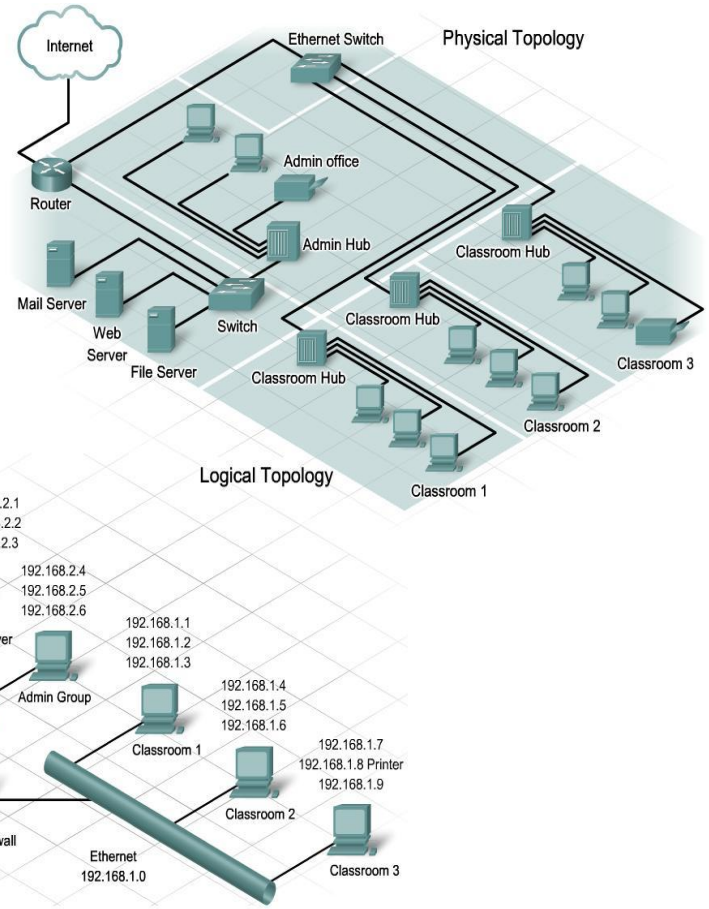
- How network devices are physically connected
- Layout of the network

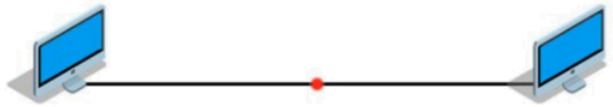
- Logical Topology

- The way signal passes through the network

- Hybrid Topology

- Combination of physical and logical



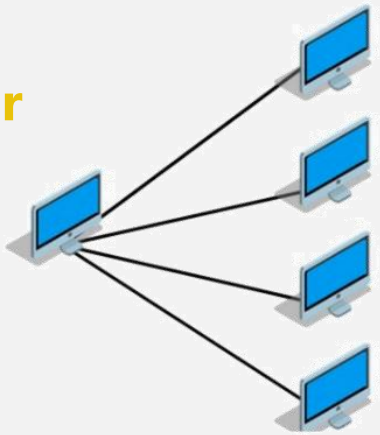


Point-to-Point (PTP, P2P)

- Foundation of all other topologies

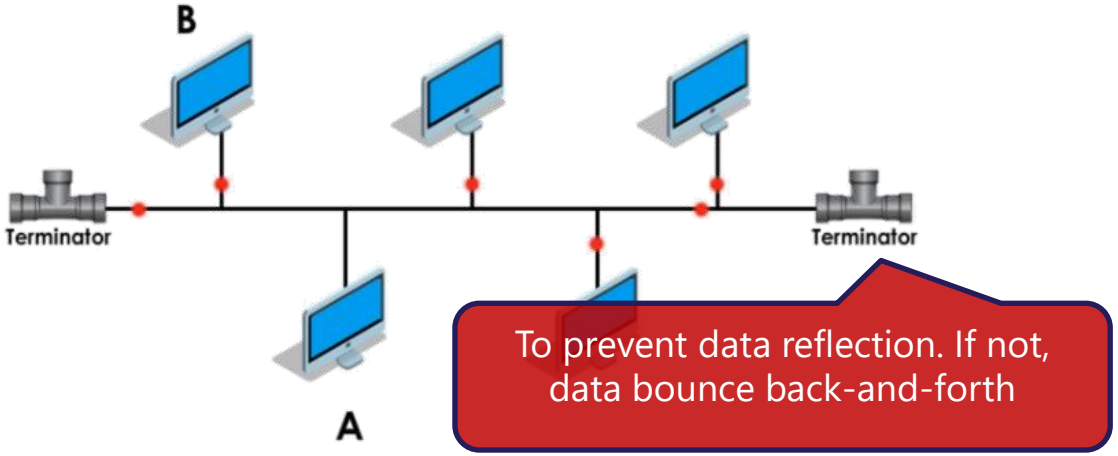
Point-to-Multipoint (PTMP, P2MP)

- Central **base station**, supports **subscriber stations**
- i.e. *Wireless LAN*



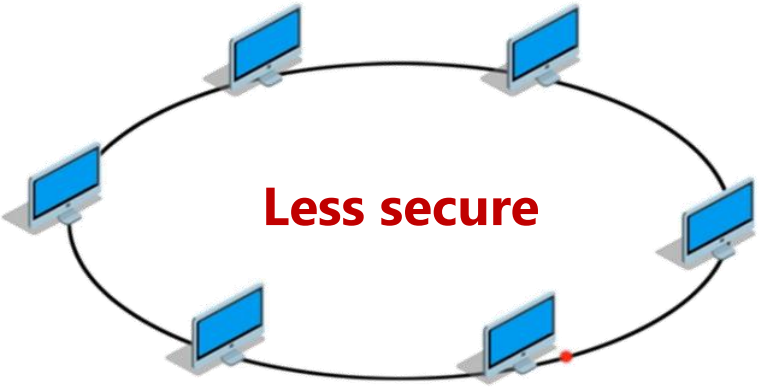
Data from **central base station** is **broadcasted;**

Data from **subscribers only** received by **central base station**



Bus (daisy chain topology)

- No central base station
- Nodes attached to **one shared bus cable**
- One single point failure=whole network failure
- Less secure

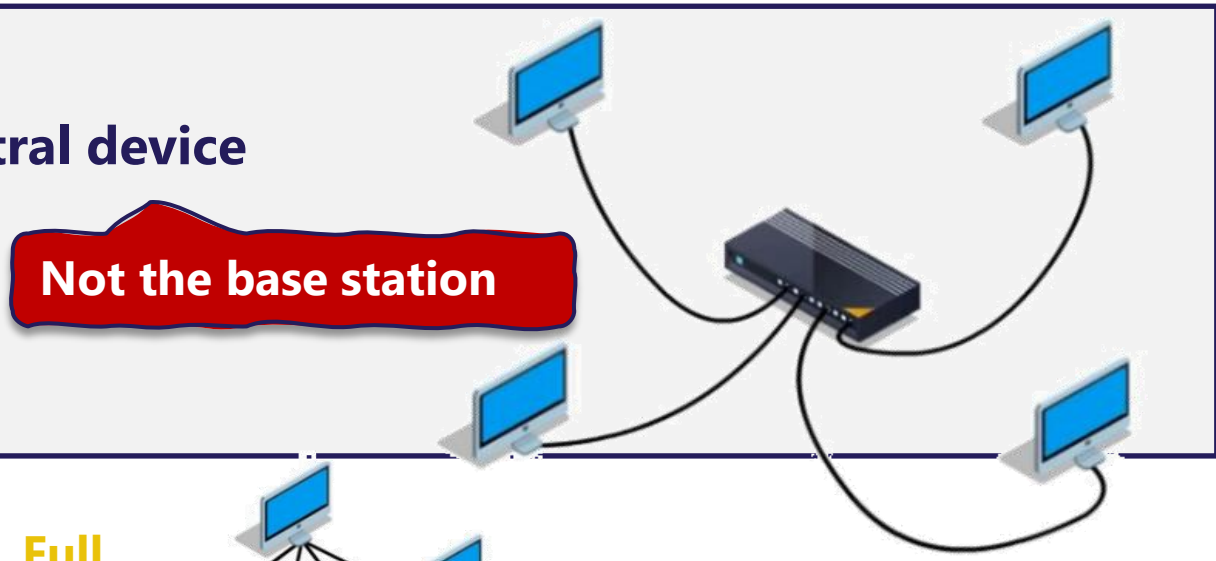


Ring Topology

- Nodes attached to a cable forming a closed loop
- No need for Terminators
- **Data** travels from **one node to another** via a token passing
- One single point failure=whole network failure

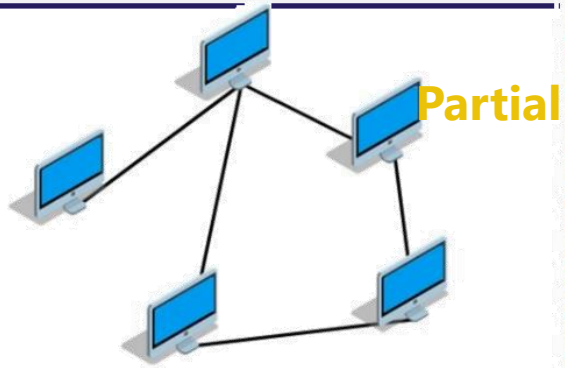
Star Topology

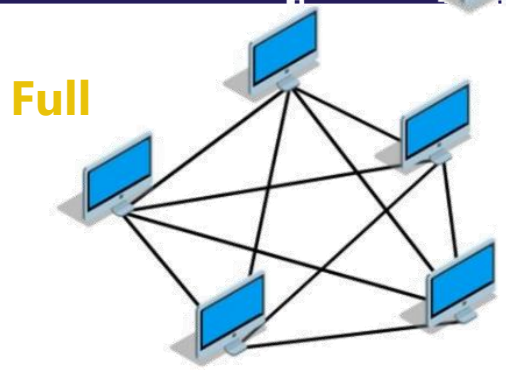
- Nodes communicate with other Nodes via a **central device**
- Fault Tolerant (if one node goes down)
- Scalable, Easy Maintenance
- **-VE: Central Point of Failure**



Mesh Topology

- Partially Meshed
- Fully Meshed



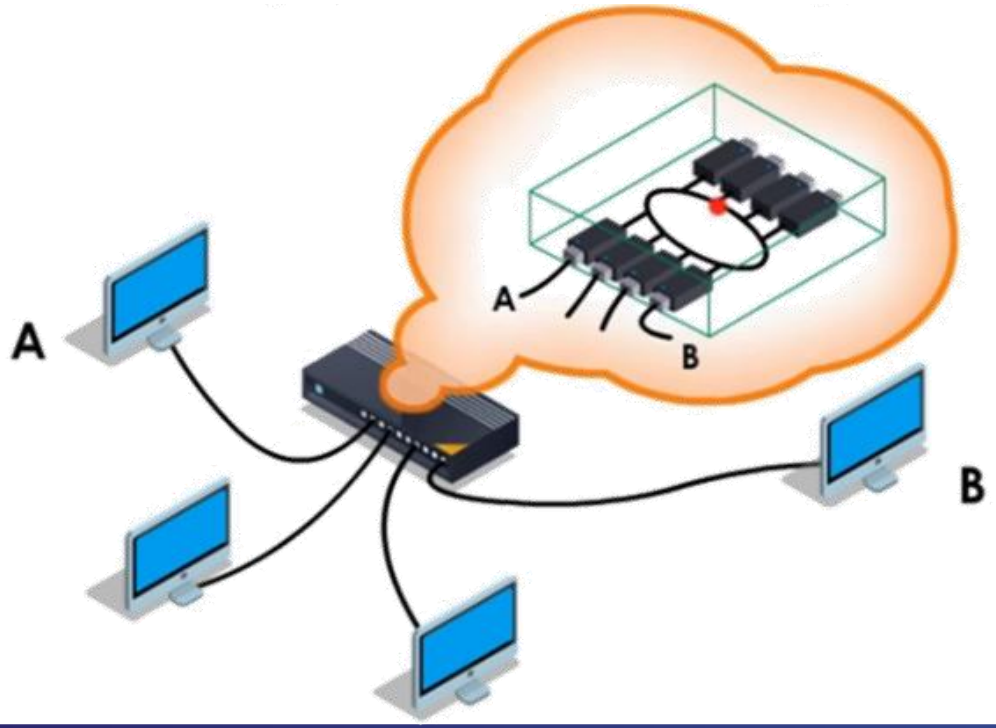


i.e. ad hoc wireless LAN

Hybrid Topology

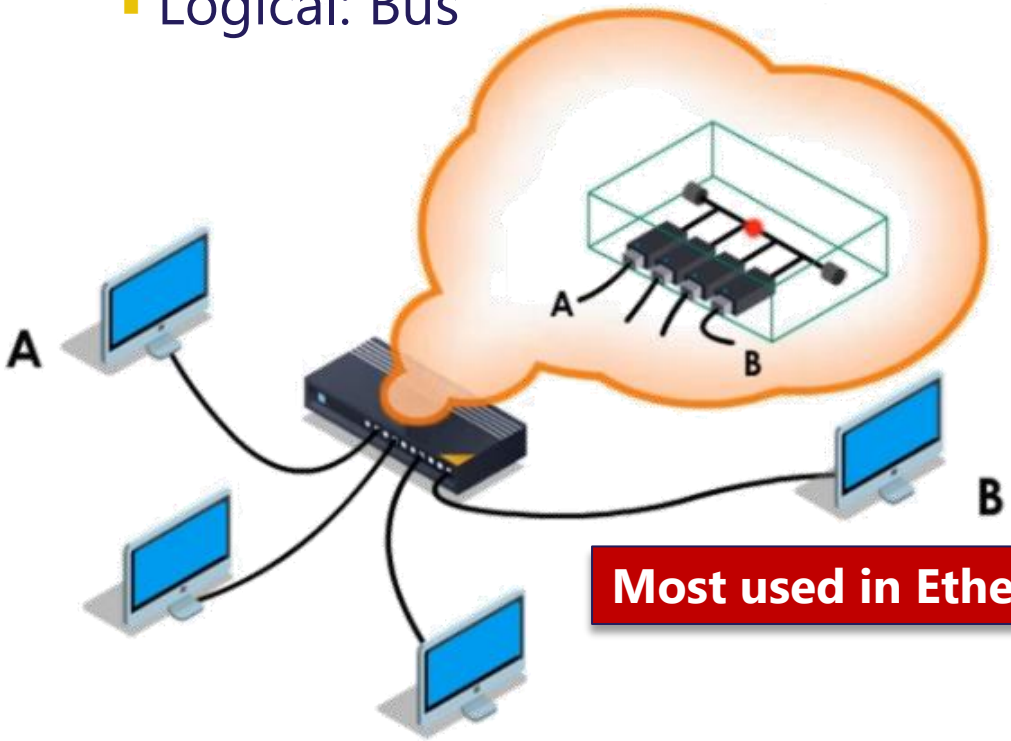
Star-ring Topology

- Physical: Star
- Logical: Ring



Star-bus Topology

- Physical: Star
- Logical: Bus



Most used in Ethernet

Transmission Media

- Transmission Modes
- Guided Media
- Unguided Media
- Wireless Signals
- Wireless Signal Attenuation

Transmission Modes

- **Simplex Mode:** communication is unidirectional
  - ✓ Sender -> Receiver | **not Receiver -> Sender**
  - ✓ Entire bandwidth can be used
    - E.g. A keyboard -> monitor
    - E.g. Radio station -> audience

Only 1-way

- **Half Duplex Mode:** communication is two-directional
  - ✓ Sender -> Receiver | Receiver -> Sender (one at a time)

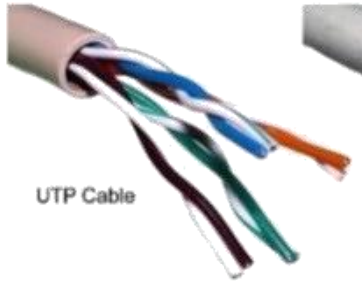
1-way at a time

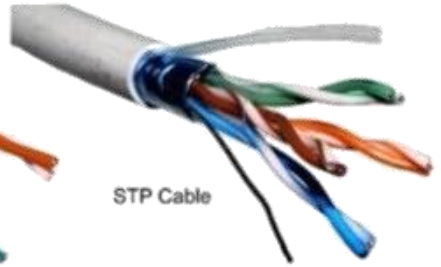
- **Full Duplex Mode:** communication is bi-directional
  - ✓ Sender -> Receiver | Receiver -> Sender (at the same time)

2-way, same time

Transmission Media

- **Guided:**
  - ✓ Twisted Pair
  - ✓ Coaxial Cable
  - ✓ Fiber Optics
- **Unguided:**
  - ✓ Radio
  - ✓ Microwave
  - ✓ Lightwave
  - ✓ Infra-red





Unshielded Twisted Pairs (UTP)

Shielded Twisted Pairs (STP)



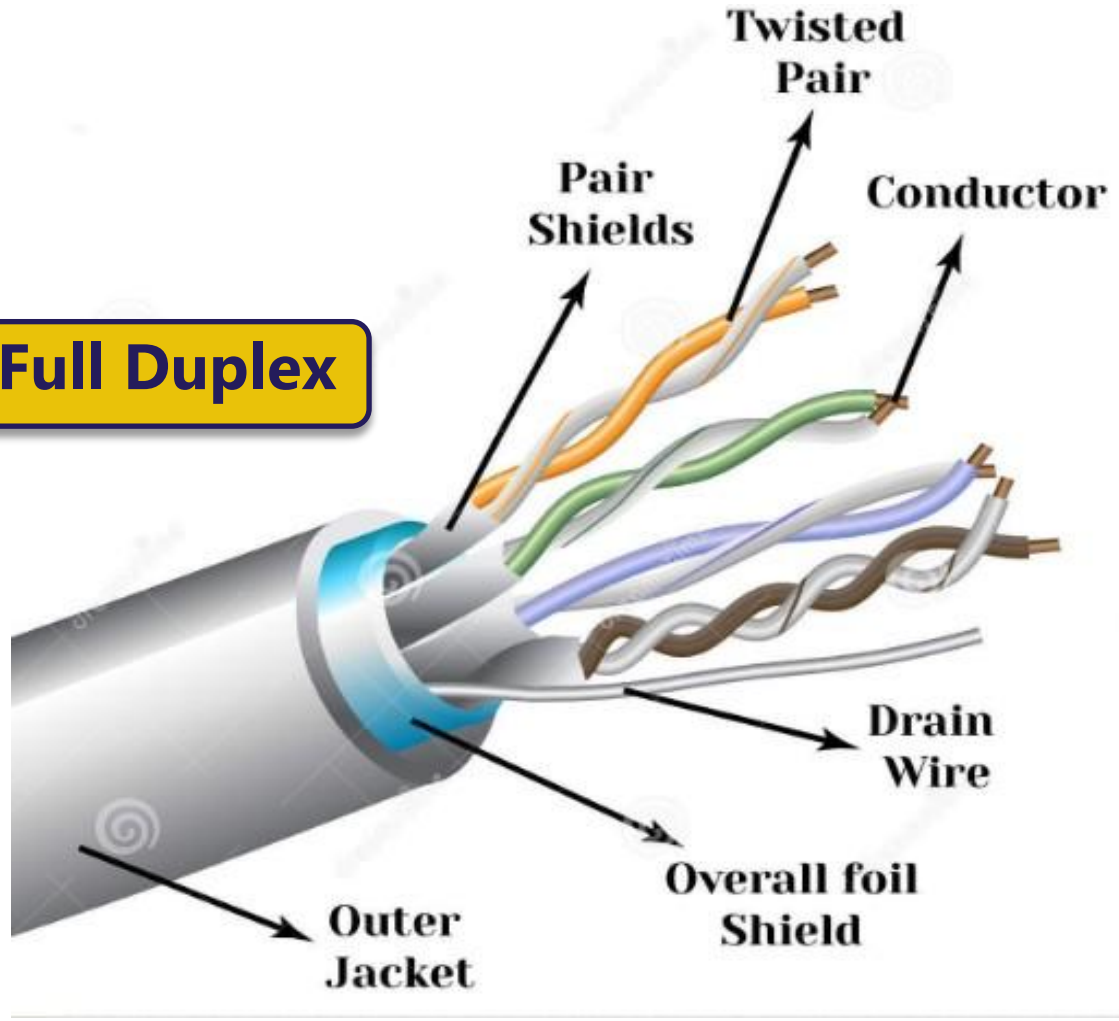
Fiber Optic Cable



Coaxial Cable

Twisted Pair

Full Duplex

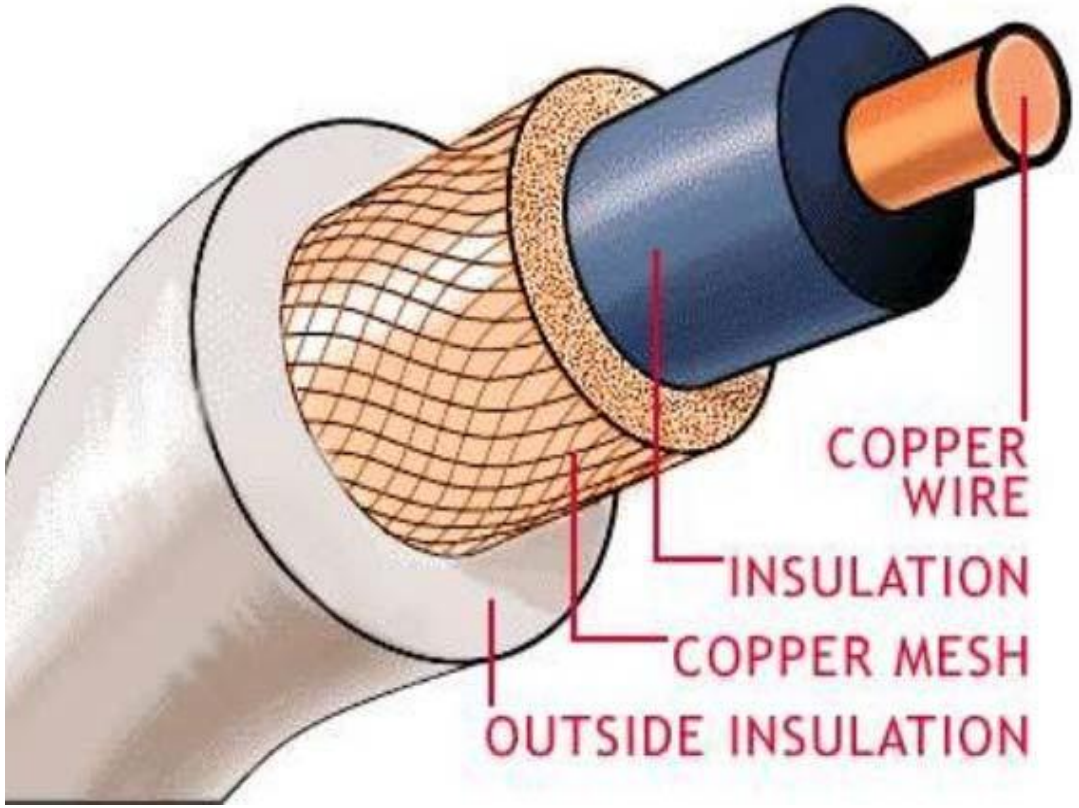


- Oldest and still **common** transmission media
- Can transmit **analog or digital** signals
- Most common is **UTP** (Unshielded Twisted Pair)
- Uses **repeaters** for long distance connection
- Less expensive

Coaxial Cable

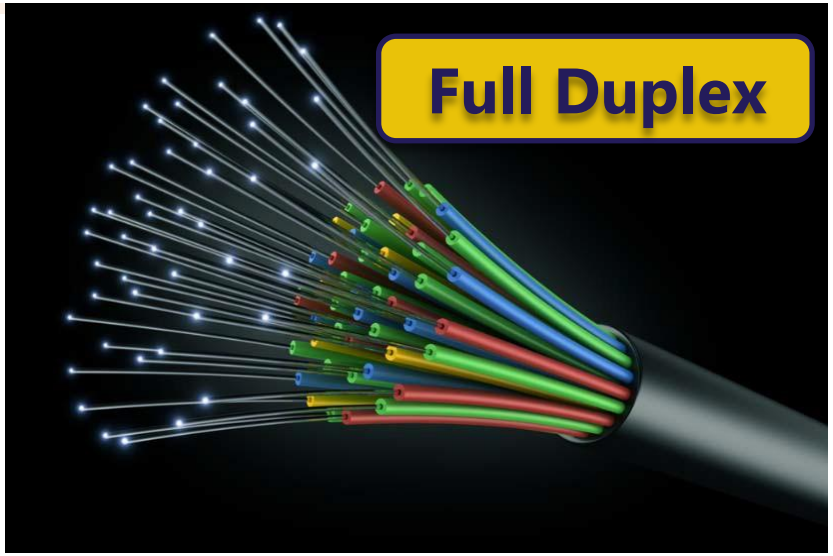
Half Duplex

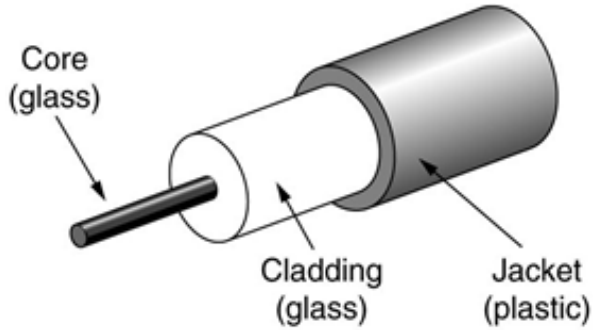
- **Better shielding** than twisted pairs
- **Not flexible**
- Two kinds:
  - ✓ 50-ohm cable (digital transmission)
  - ✓ 76-ohm cable (analogy transmission & cable television)

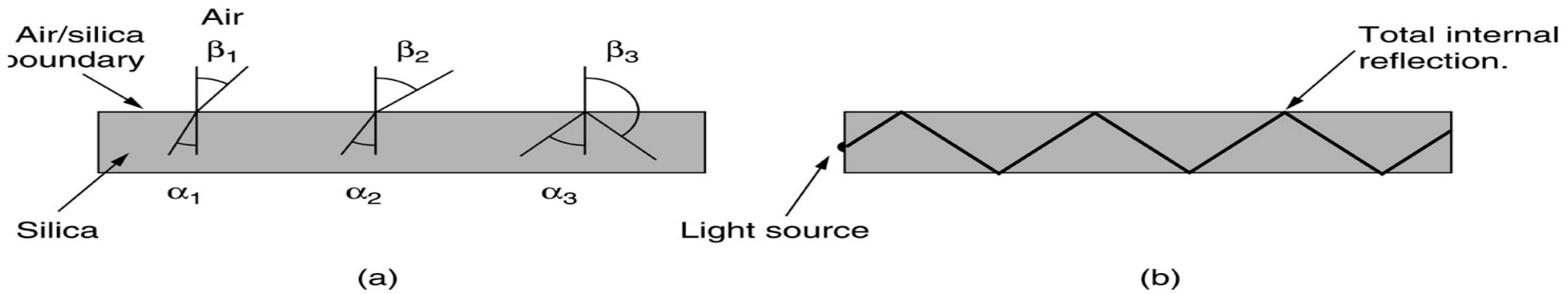


Fiber Optic

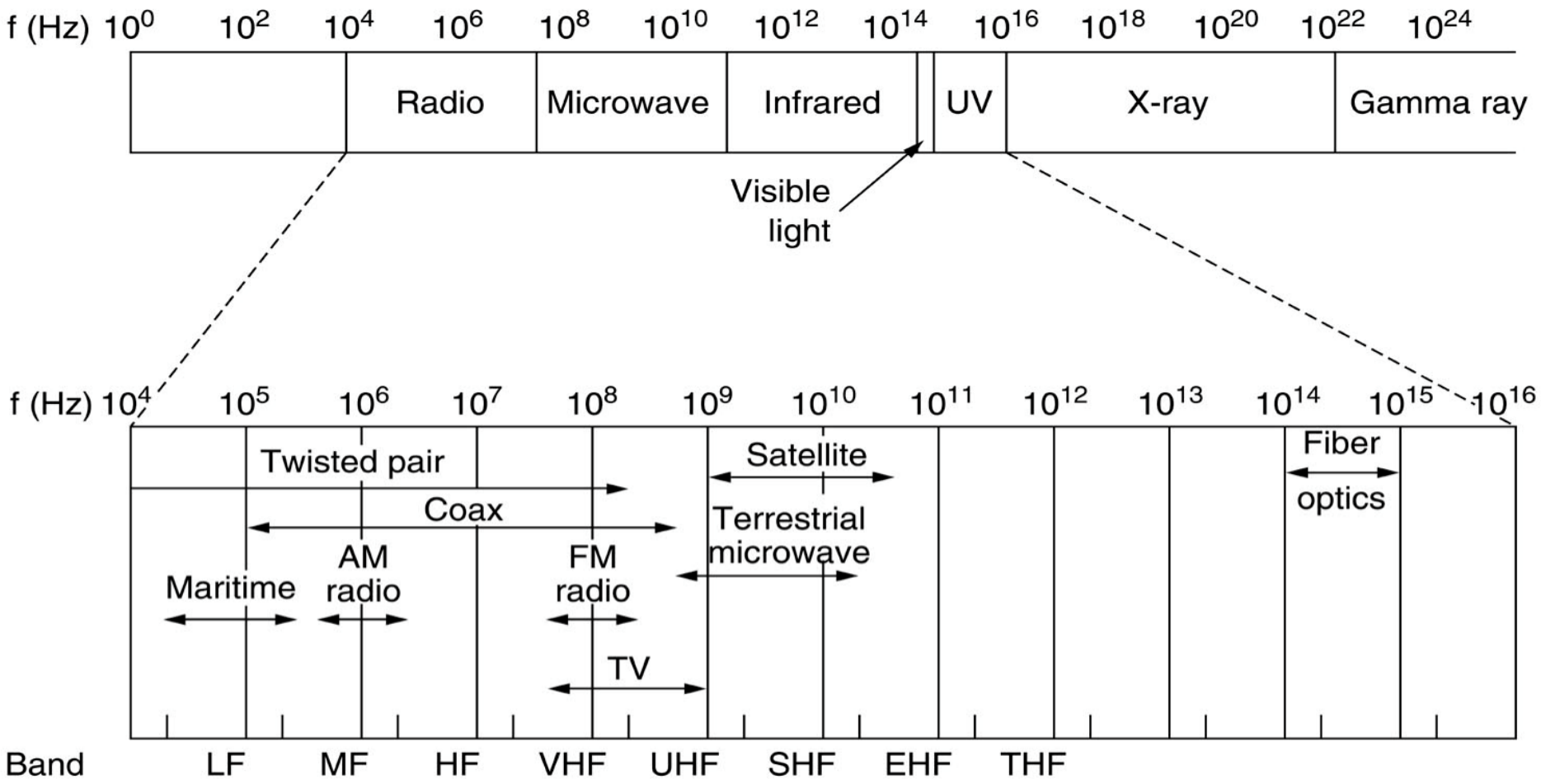
- Light source (speed of light)
- Transmission medium (ultra-thin fiber of glass)
- Detectors





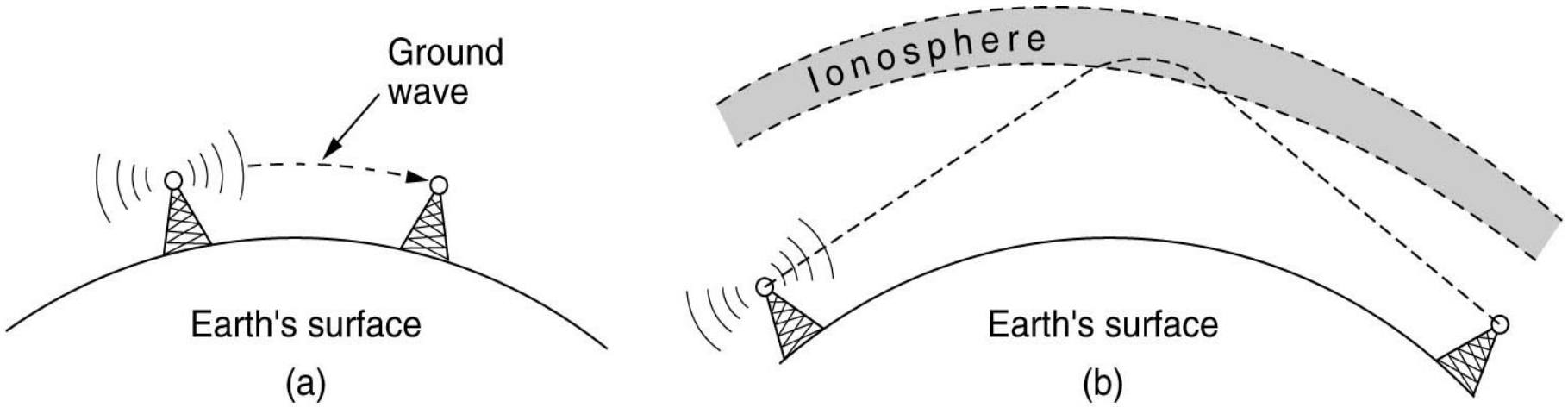


Unguided: Electromagnetic Spectrum



Radio Transmission

- Easy to generate & travel long distance
- Omni-directional





*(a) VLF, LF & MF bands, radio waves **follow the curvature** of the earth.*

*(b) HF band **bounced off** the ionosphere. (Tanenbaum)*

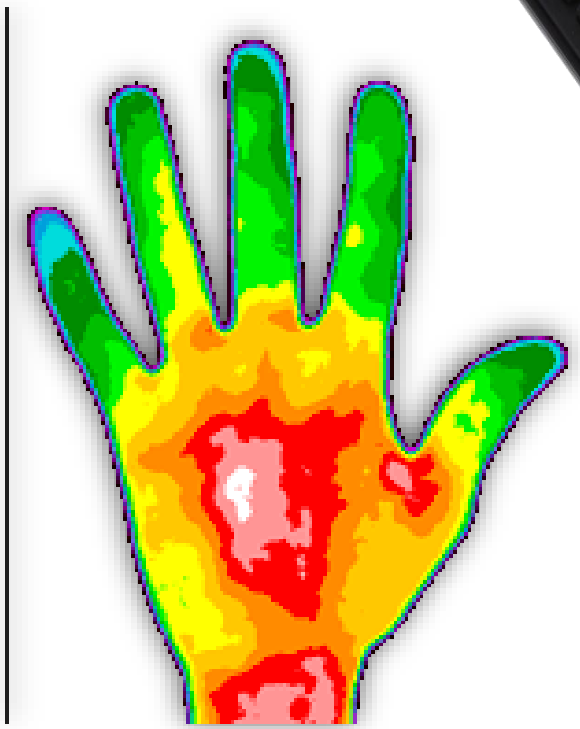
Microwave Transmission

- Travel in **straight line**
- Used for:
  - ✓ Long-distance telephone communication
  - ✓ Television distribution
  - ✓ Etc.



Infrared & Millimeter Waves

- **Short-range** communication
  -Remote controls
- **Cheap** and easy to built
- **Cannot penetrate** solid objects



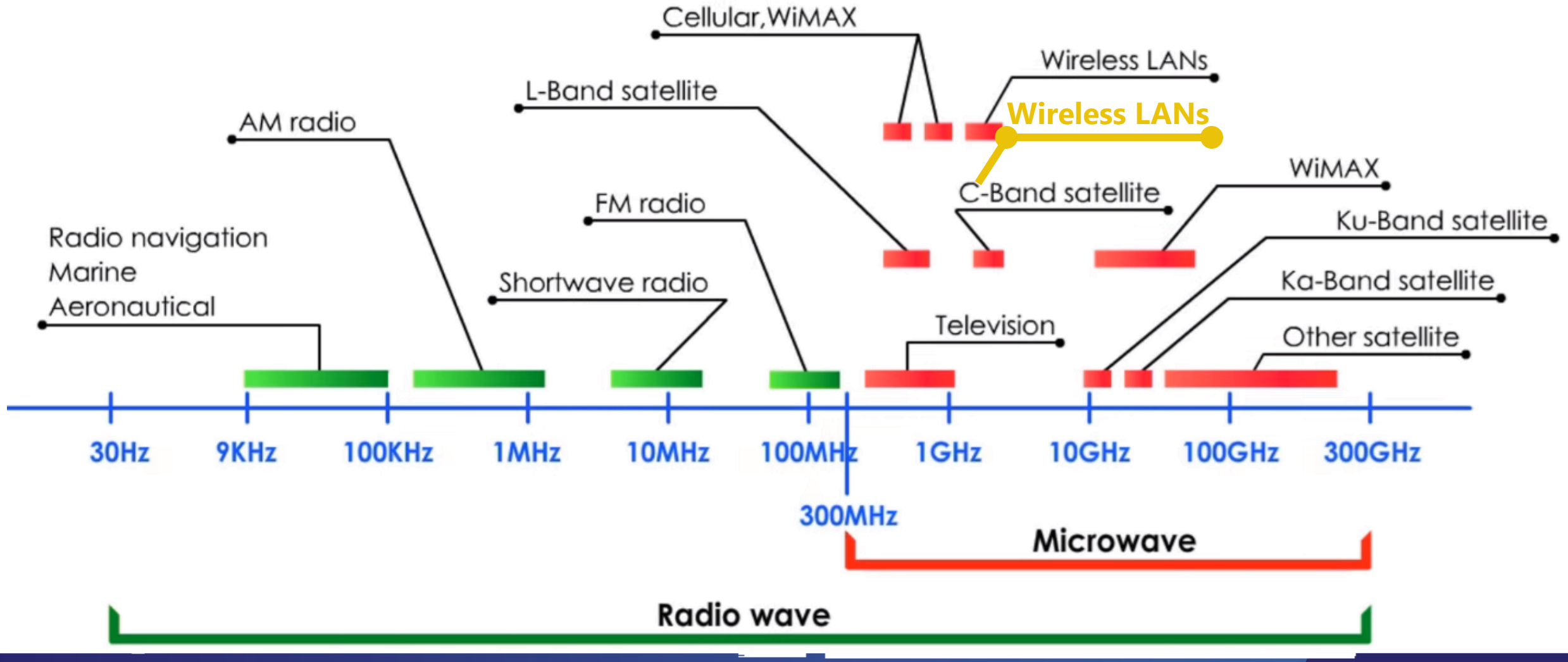


Light Wave Transmission

- **Laser or Light**
- **Unidirectional**
- Offer **high bandwidth** & very low cost
- **Cannot penetrate** rain or thick fog



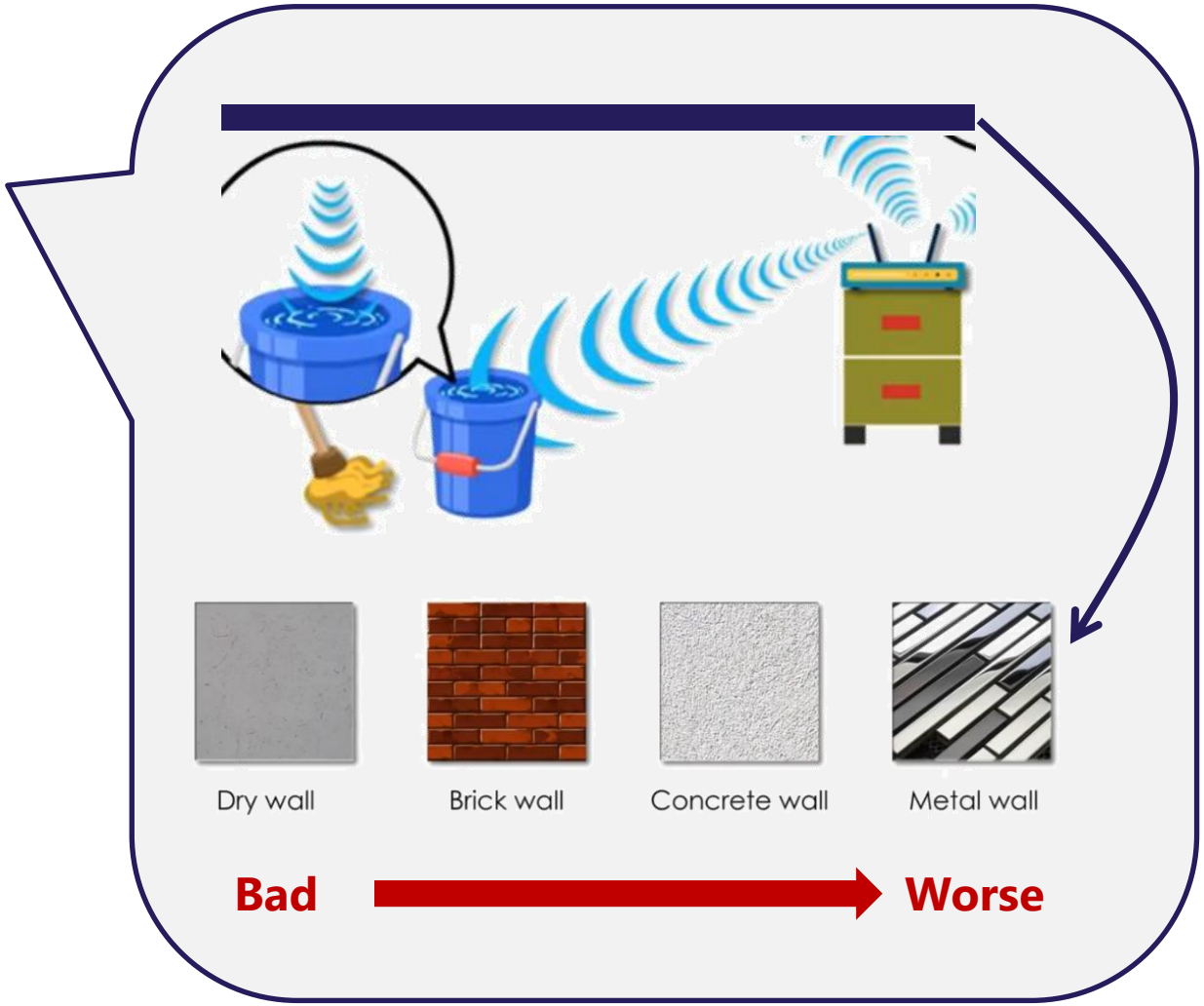
Wireless Signals



Wireless Signal Attenuation?

- **Absorption**
  - Some materials absorb signal's strength
- **Reflection**
  - Some absorbed, some reflected
  - Not necessarily bad







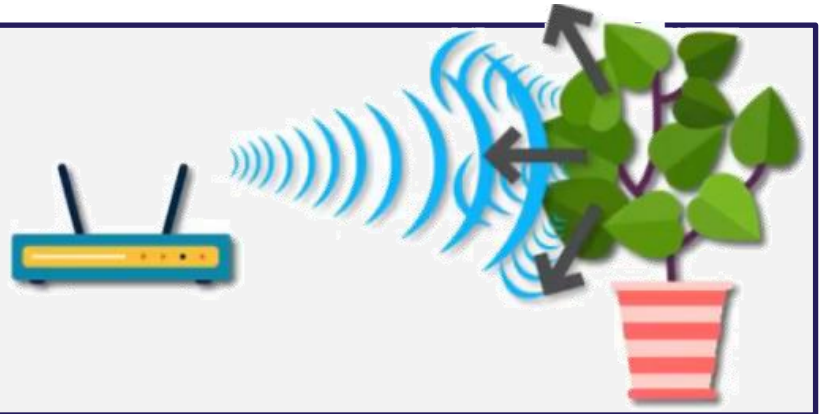
Diffraction

- Sharp edges, corners, etc.



Scattering

- Diffusion of wireless signal at an object's uneven surface
- i.e. plant leaves, **dust**, **smoke**, or micro **drops of water**







Interference

- Devices operating on same 2.4Ghz

Last Mile Technologies

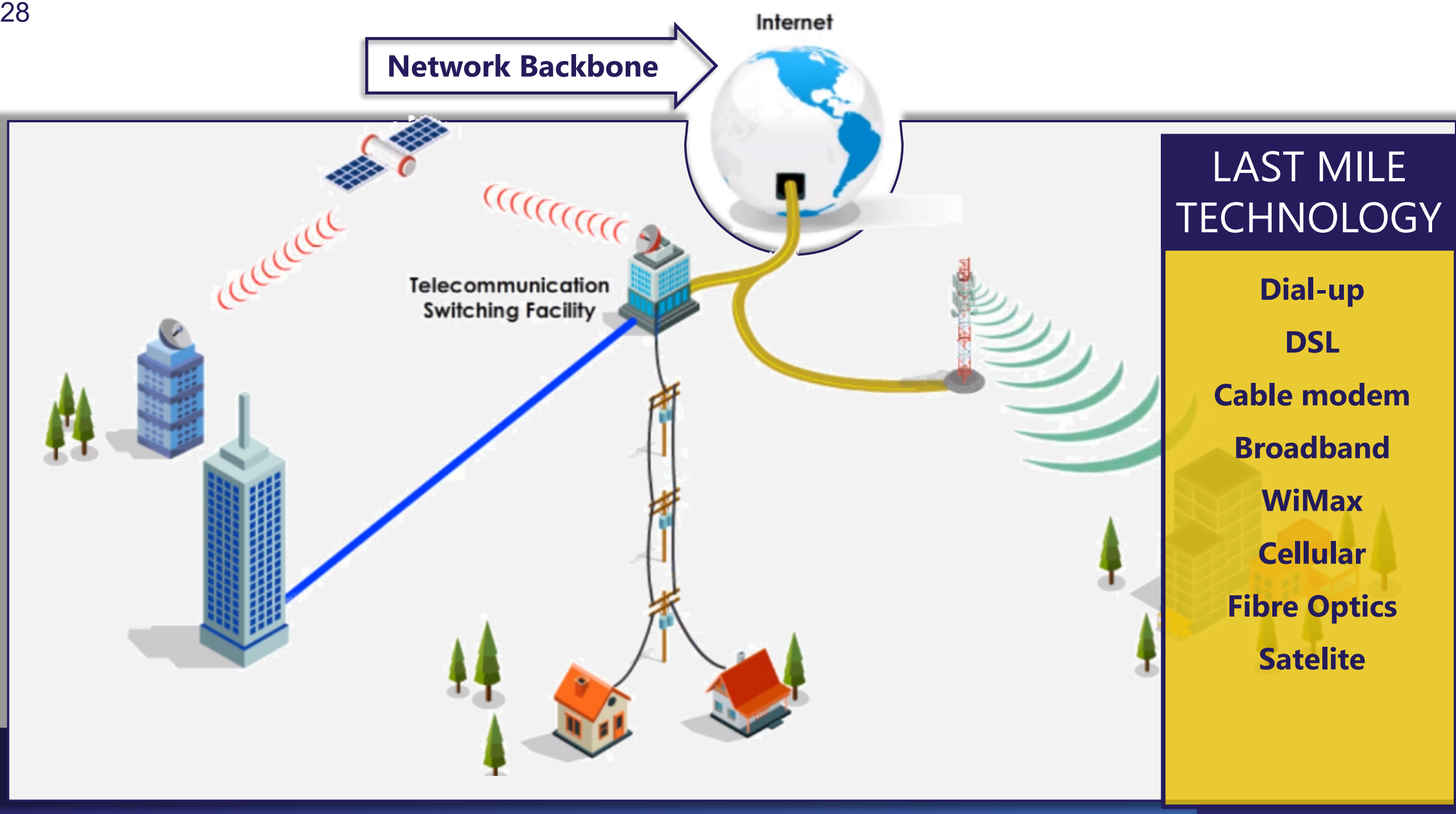
- Dial-up
- ISDN
- DSL/ADSL
- NBN

Last Mile Technologies

- Refers to any **telecommunication methods, devices** and **media** that **carry signals over the last/first mile**
- **Typically the speed bottleneck**

E.g. Network backbone use Fibre Optics but the last mile still use copper wires



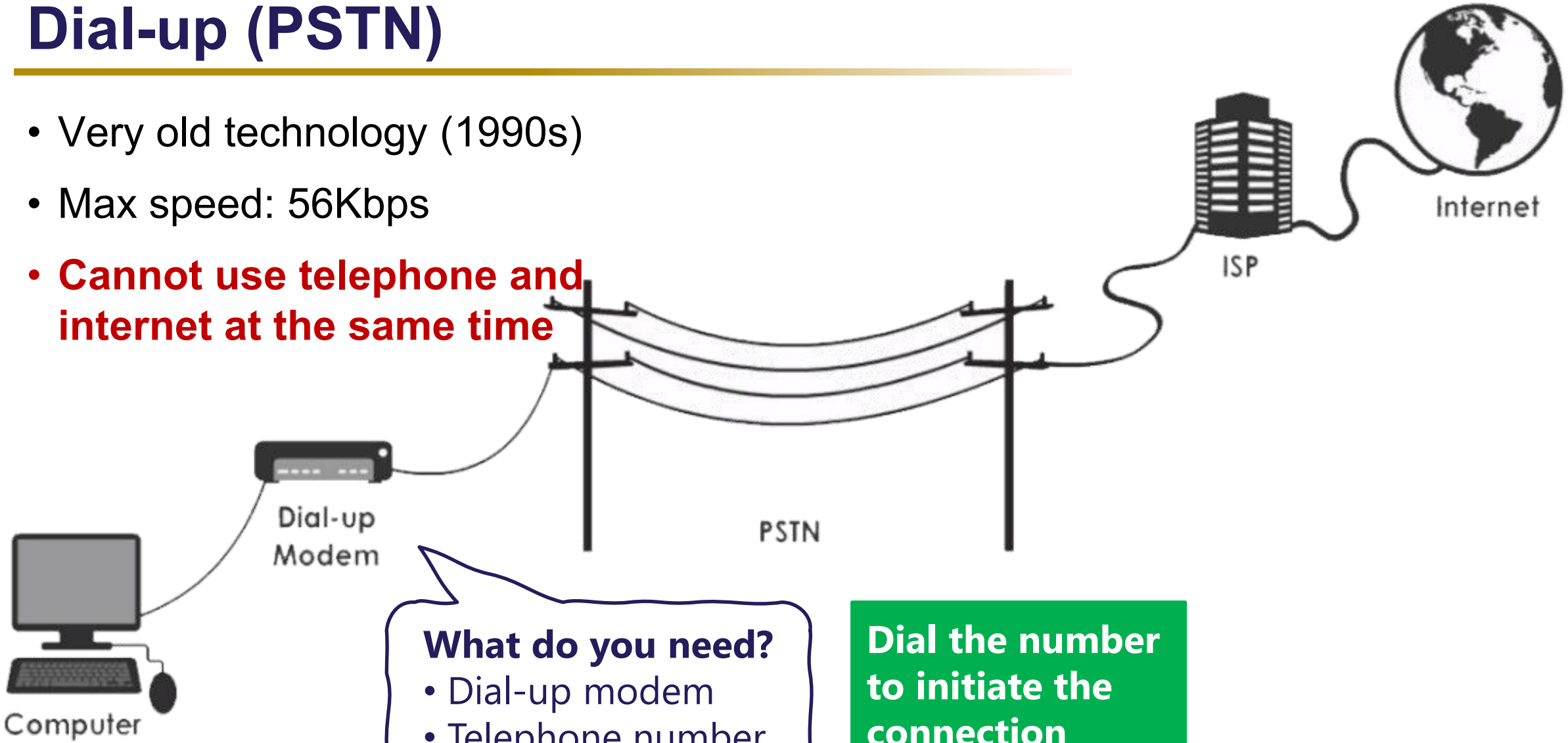


LAST MILE TECHNOLOGY

- Dial-up
- DSL
- Cable modem
- Broadband
- WiMax
- Cellular
- Fibre Optics
- Satelite

Dial-up (PSTN)

- Very old technology (1990s)
- Max speed: 56Kbps
- **Cannot use telephone and internet at the same time**



What do you need?
- Dial-up modem
- Telephone number

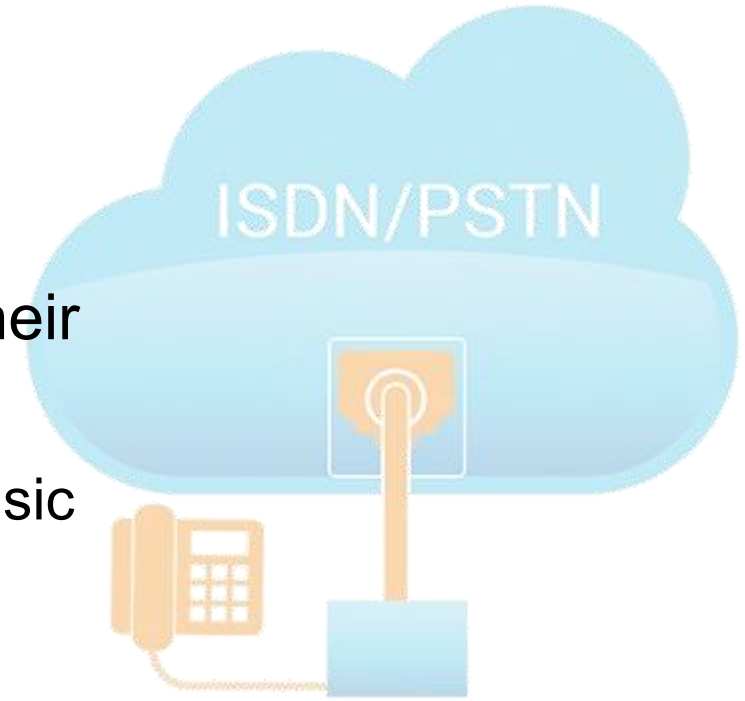
Dial the number to initiate the connection

ISDN

- Integrated Services Digital Network (ISDN) provides **digital transmission of voice and data** services

- ISDN supports **multiples channels** for voice and data.

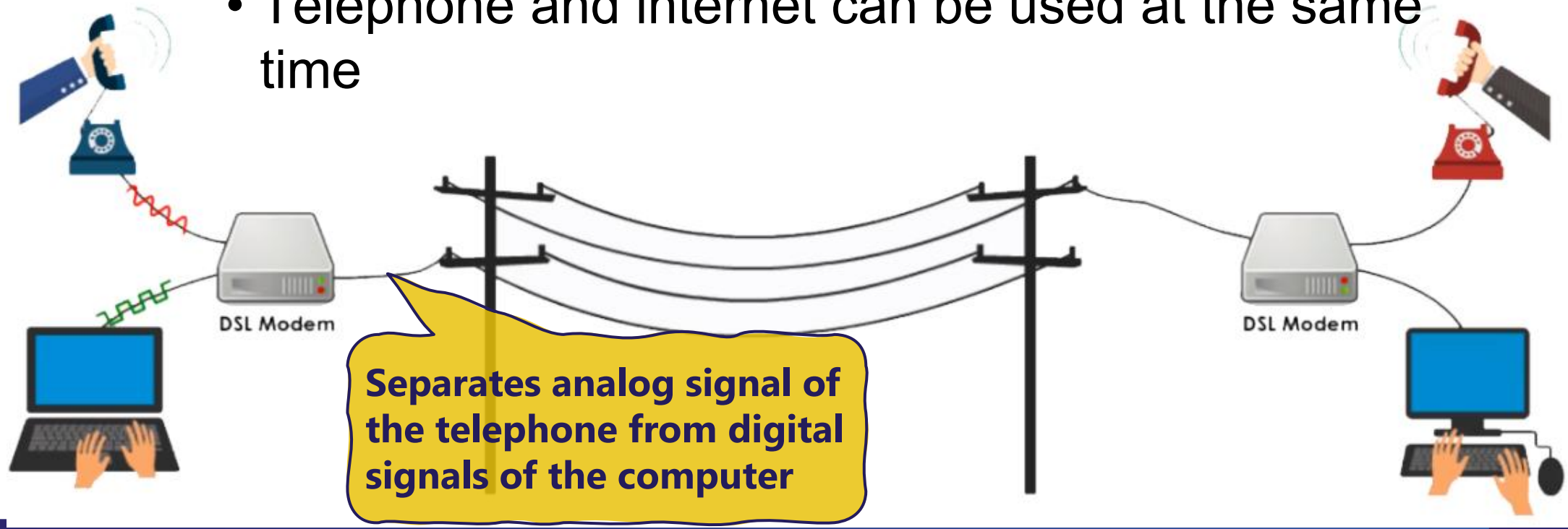
- Medium to Large Businesses: Option of integrating with their phone systems (PABX) to enable **multiple features**
  - ✓ Like using a 100-number range, groups, queues, on hold music and RVAs, etc.



With **NBN** (National Broadband Network), PSTN and ISDN are phased out.

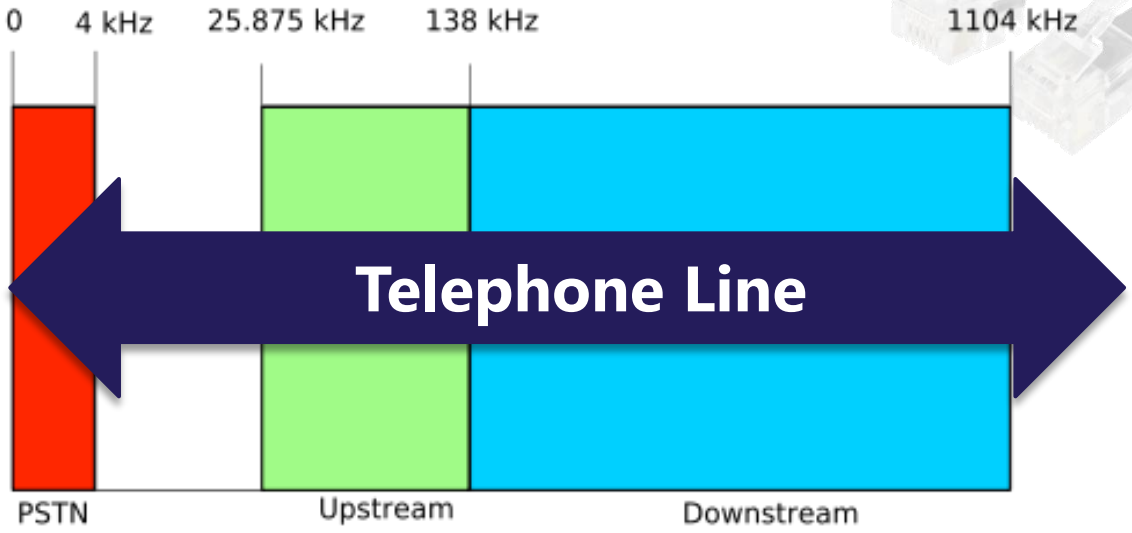
Digital Subscriber Line (DSL)

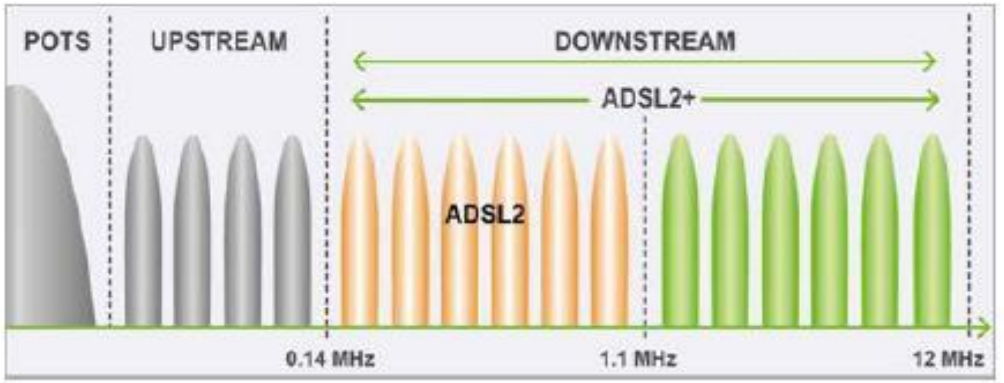
- Faster data transfer over PSTN
- **Always on** connection (No dialing required)
- Telephone and internet can be used at the same time



DSL

- **ADSL (Asymmetric DSL)**
  - -Downstream is faster than upstream
- **For business use:**
  - **-SDSL (Symmetric DSL)**
    - -i.e. video conferencing
  - **-HDSL (High bit-rate DSL)**
  - **-VDSL (Very high bit-rate DSL)**

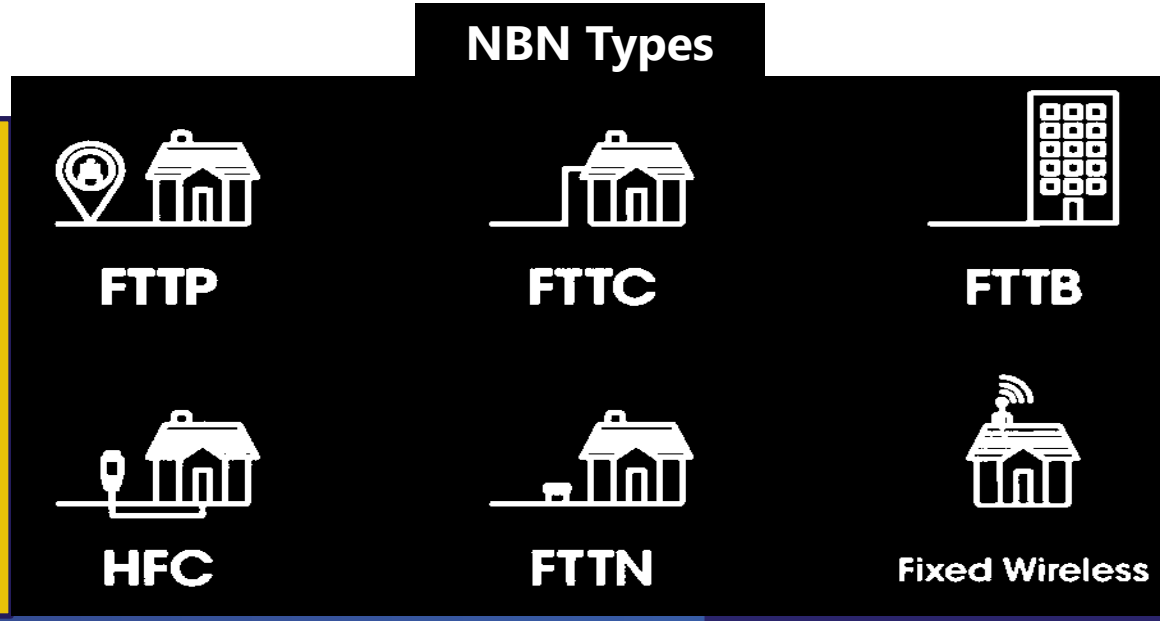




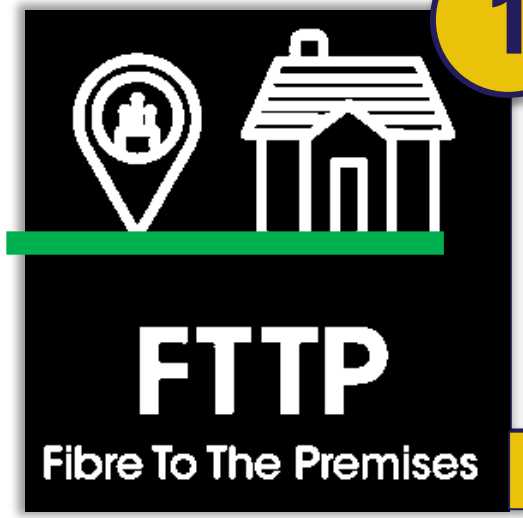
National Broadband Network (NBN)

- National network of communication infrastructure currently being built on behalf of federal government
- **For internet and phone *(VoIP)***
- Aims to deliver network speed of **50Mbps – 1Gbps**

- Broadband Network via Fibre Optics, Fixed Wireless, Satellite Technology for home users
  - **Always on** connection *(no dialing)*



NBN Types



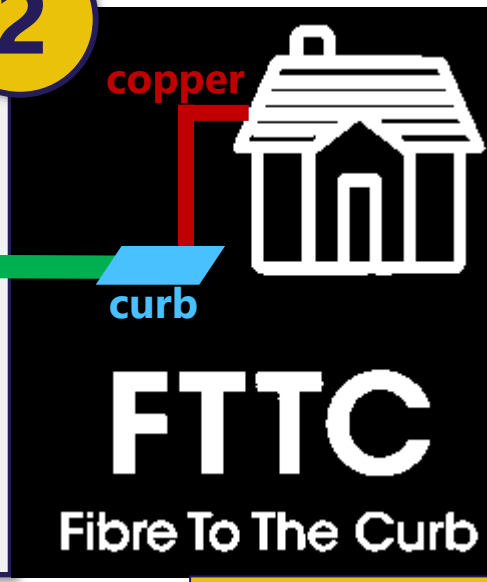
1

- Fibre is connected all the way from exchange to office/house
- **GOLD** standard in NBN

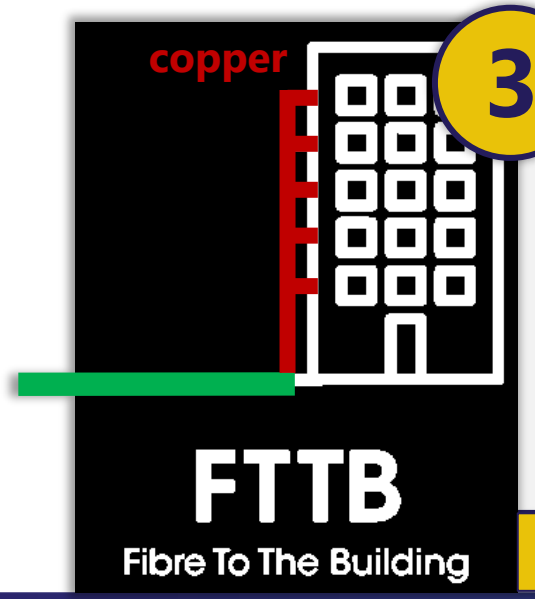
<= 1 Gbps

2

- Fibre to curb or driveway
- Existing copper lines from curb to premises



<= 100 Mbps

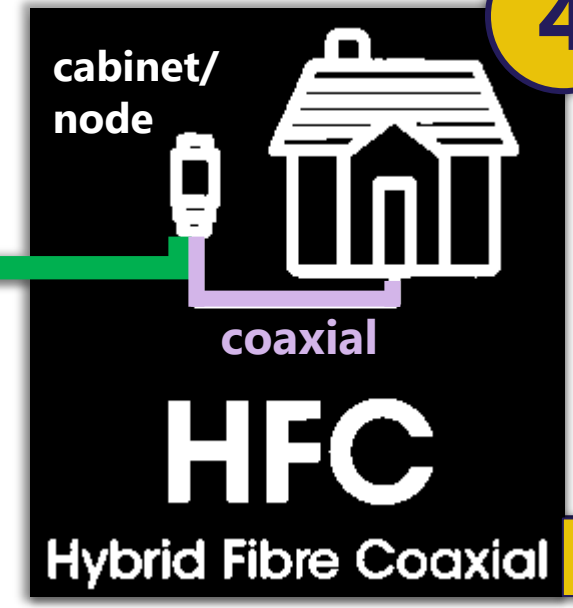


3

- Fibre to central point in the building
- Copper to each socket

<= 100 Mbps

NBN Types



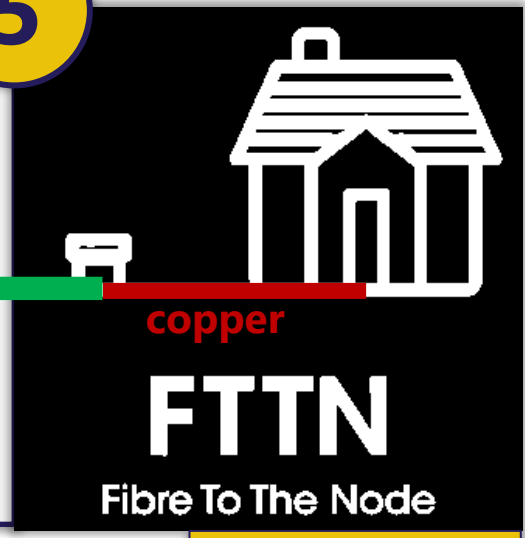
4

- Fibre to cabinet (a.k.a node) near your premises
- Existing coaxial cable from cabinet to premises

<= 100 Mbps

5

- Fibre to node near your premises
- Existing copper lines (50m-1.5km) from node to premises



25-100 Mbps



6

- R e areas
- Fixed antenna to receive the signal from the tower
- Wireless can be affected by many factors

12-50 Mbps

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Thank you

